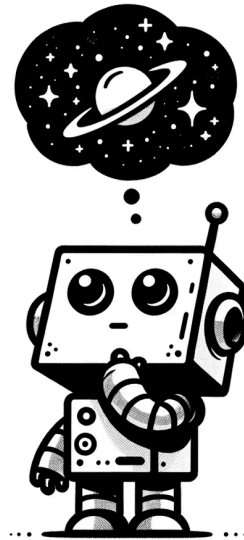


Reinforcement Learning: Classical Foundations and the LLM era



Carol(in) Cuesta-Lazaro

Flatiron Institute

TD-GAMMON



AlphaGo



AlphaGo beat
SciNews



ChatGPT (RLHF)

Prompt
Explain the moon landing to a 6 year old in a few sentences.

Completion
GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

1992

2013

2016

2019

2022

2025

DQN



DQN Breakout
Google DeepMind



0:00 / 2:58



Hide And Seek

Multi-Agent Hide and Seek
OpenAI



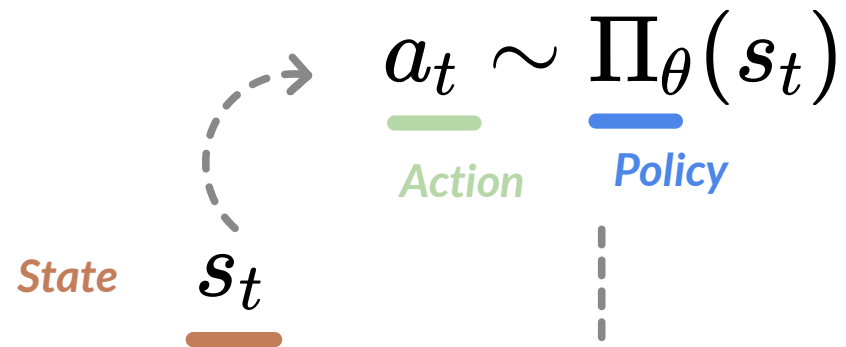
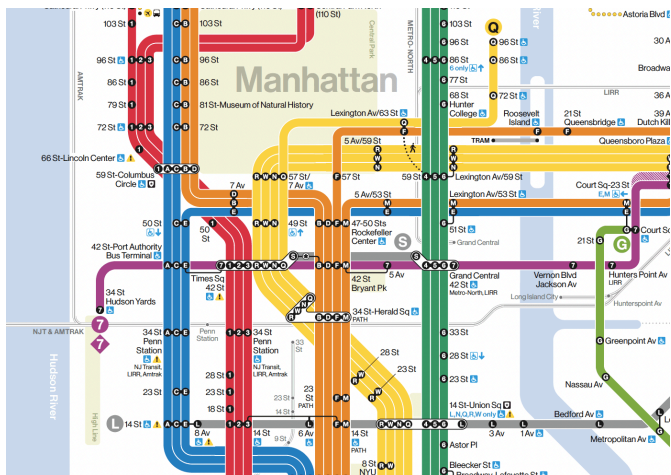
Reasoning (RLVR)



Agent

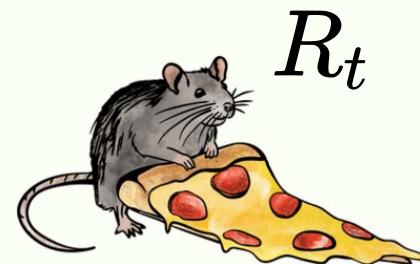


Environment



s_{t+1}

Reward



RL vs Supervised Learning

- In RL the **data distribution is changing**

The agent generates its own training data by acting

- Trade off between **exploration and exploitation**

Should I do what's worked before, or try something new?

- Reward signal is **evaluative, not instructive**

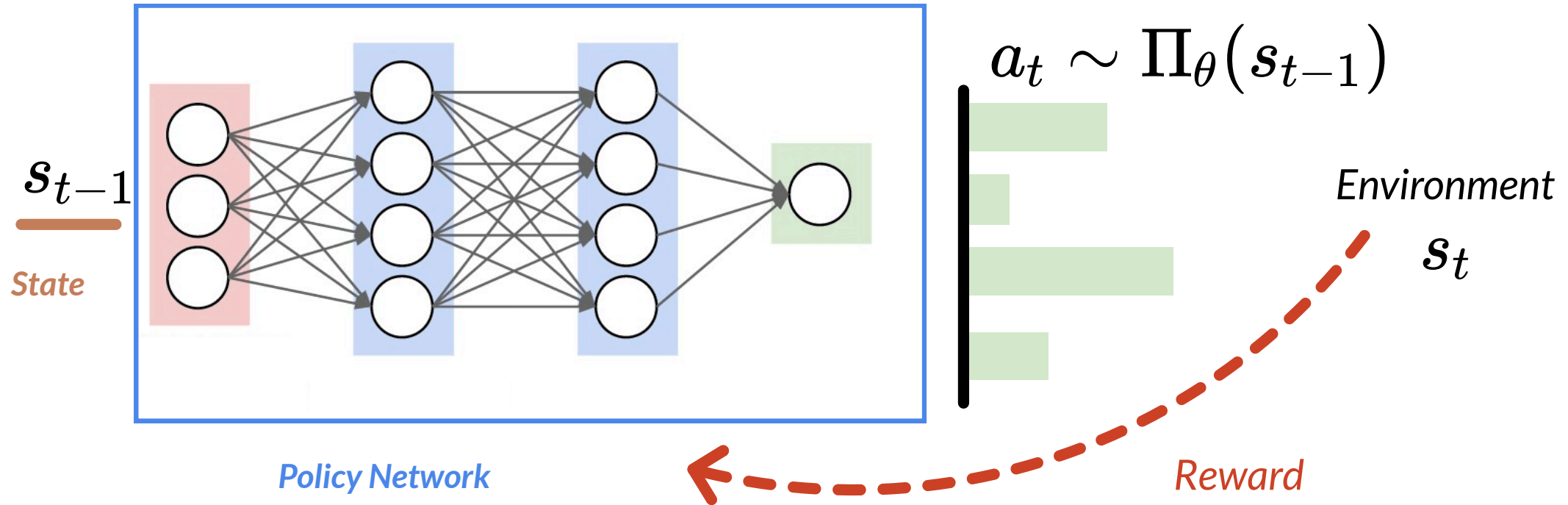
Tells you how good, not what was right

- **Delayed Feedback**

The consequence of an action may only show up many steps later

- Very **stochastic** outcomes

The Learning Problem



$$\frac{\partial R}{\partial \theta} = \frac{\partial R}{\partial s_T} \cdot \frac{\partial s_T}{\partial a_{T-1}} \cdot \frac{\partial a_{T-1}}{\partial s_{T-1}} \cdots \frac{\partial a_0}{\partial \theta}$$

Not Differentiable!

Reinforcement Learning from Human Feedback (RLHF)

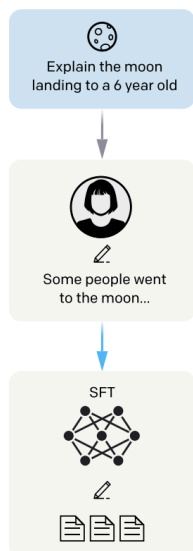
Step 1

Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3 with supervised learning.



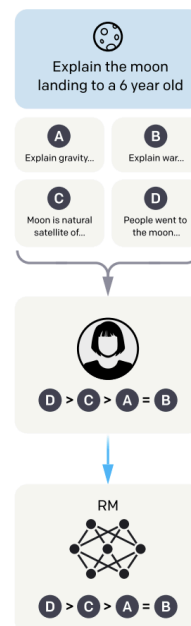
Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

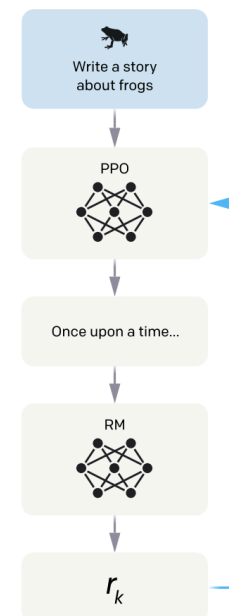
Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



Reinforcement Learning from Verifiable Rewards (RL VR)

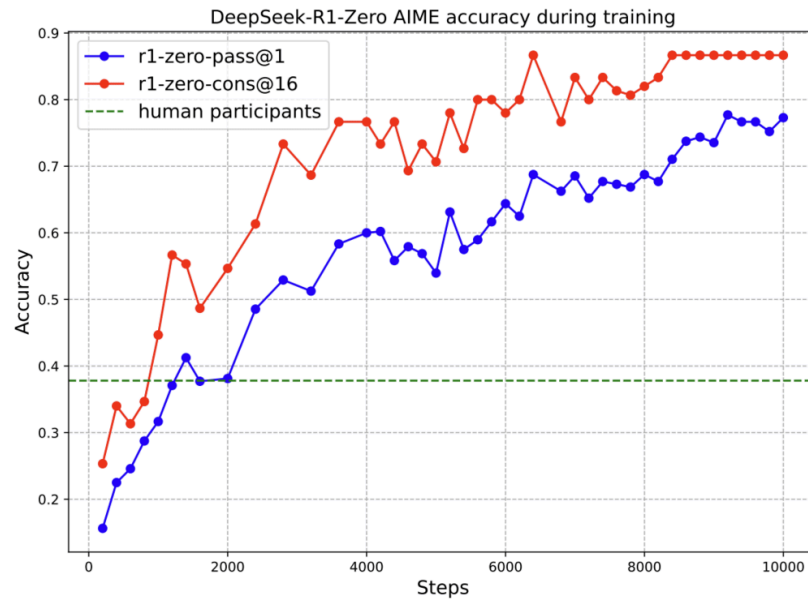


Figure 1 | (a) AIME accuracy of DeepSeek-R1-Zero during training. AIME takes a mathematical problem as input and a number as output, illustrated in Table 32. Pass@1 and Cons@16 are described in Supplementary D.1. The baseline is the average score achieved by human participants in the AIME competition. (b) The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time. Note that a training step refers to a single policy update operation.

The increase in thinking time fosters the autonomous development of sophisticated behaviors. Specifically, DeepSeek-R1-Zero increasingly exhibits advanced reasoning strategies such as reflective reasoning and systematic exploration of alternative solutions (see Figure 9(a) in Supplementary C.2 for details), significantly boosting its performance on verifiable tasks like math and coding. Notably, during training, DeepSeek-R1-Zero exhibits an “aha moment” (Table 2), characterized by a sudden increase in the use of the word “wait” during reflections (see Figure 9(b) in Supplementary C.2 for details). This moment marks a distinct change in reasoning patterns and clearly shows the self-evolution process of DeepSeek-R1-Zero.

The self-evolution of DeepSeek-R1-Zero underscores the power and beauty of RL: rather than explicitly teaching the model how to solve a problem, we simply provide it with the right incentives, and it autonomously develops advanced problem-solving strategies. This serves as a reminder of the potential of RL to unlock higher levels of capabilities in LLMs, paving the way for more autonomous and adaptive models in the future.

Are models learning something new via RL?

Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?

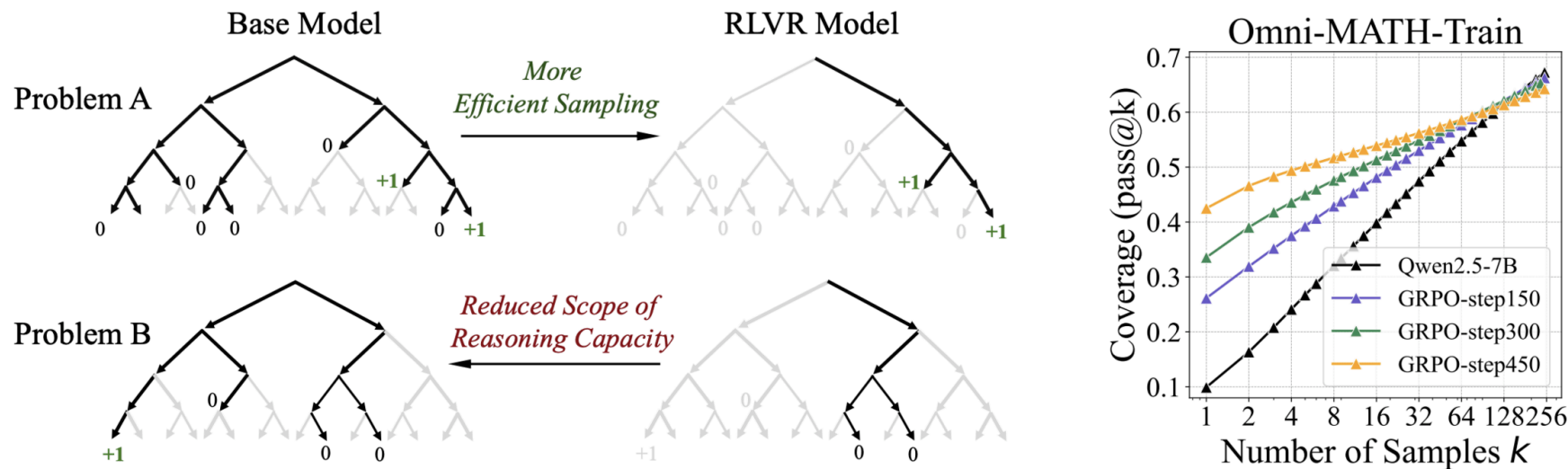


Figure 1: **(Left)** The effect of current RLVR on LLM’s reasoning ability. Search trees are generated by repeated sampling from the base and RLVR-trained models for a given problem. Grey indicates paths that are unlikely to be sampled by the model, while **black** indicates paths that are likely to be sampled. **Green** indicates correct paths, which has positive rewards. Our key finding is that all reasoning paths in the RLVR model are already present in the base model. For certain problems like Problem A, RLVR training biases the distribution toward rewarded paths, improving sampling efficiency. However, this comes at the cost of reduced scope of reasoning capacity: For other problems like Problem B, the base model contains the correct path, whereas that of the RLVR model does not. **(Right)** As RLVR training progresses, the average performance (*i.e.*, pass@1) improves, but the coverage of solvable problems (*i.e.*, pass@256) decreases, indicating a reduction in LLM’s reasoning boundary.

Are high likelihood responses more correct?

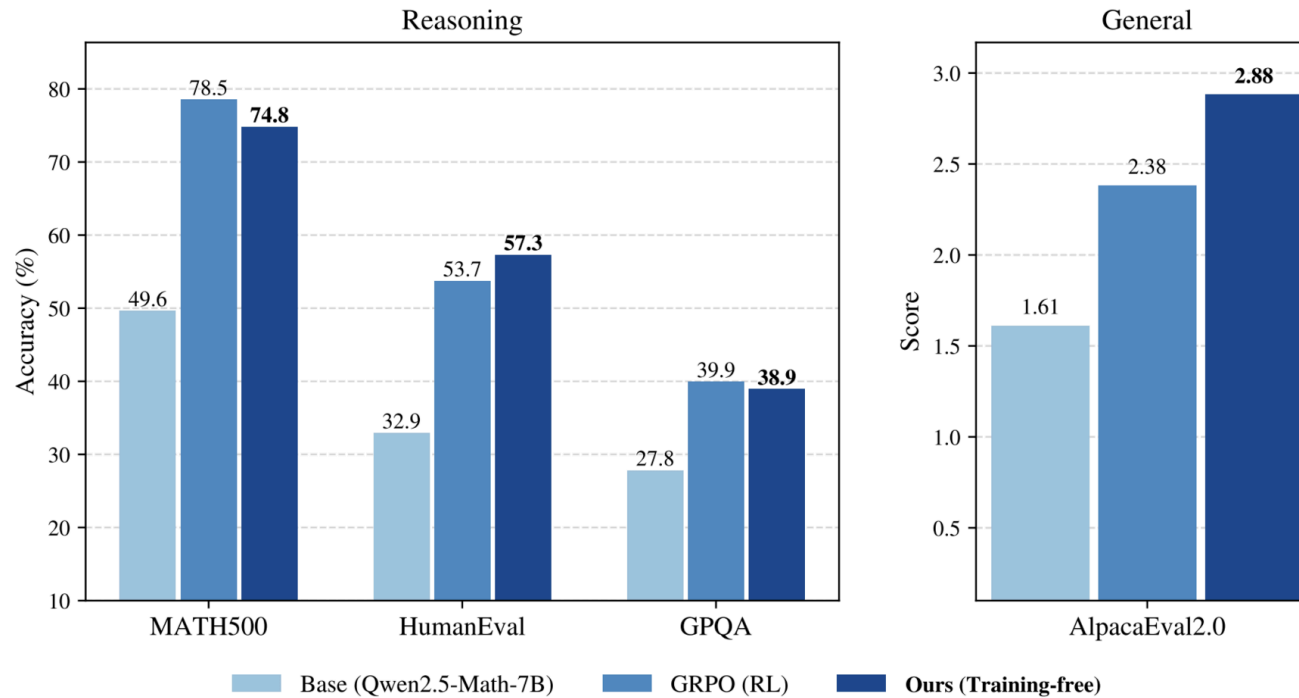


Figure 1: **Our sampling algorithm can match and outperform RL-posttraining.** Left: we compare our sampling algorithm (ours) against the base model (base) and RL-posttraining (GRPO) on three *verifiable reasoning* tasks (MATH500, HumanEval, GPQA). Right: we compare them on an *unverifiable general* task (AlpacaEval2.0). Our algorithm achieves comparable performance to GRPO within the posttraining domain (MATH500) but can *outperform* on out-of-domain tasks such as HumanEval and AlpacaEval.

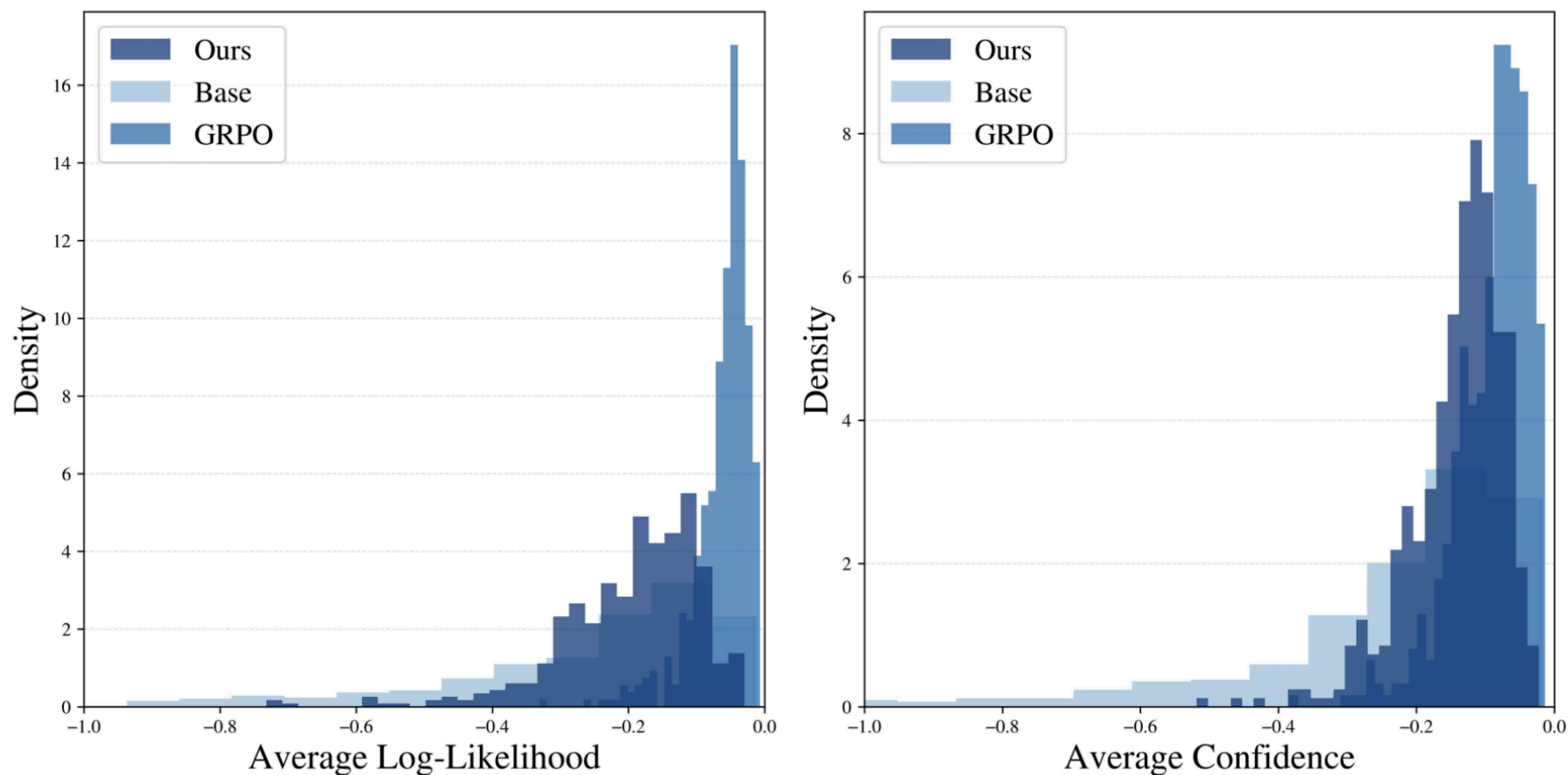
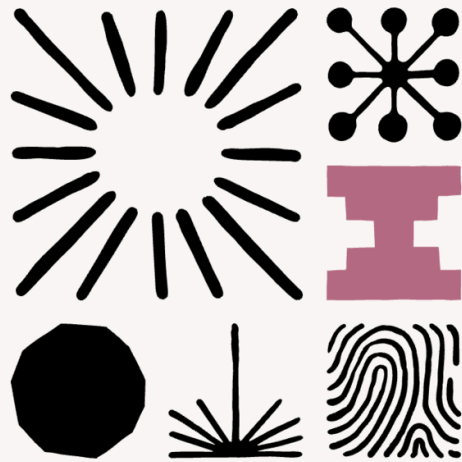


Figure 4: **Base model (Qwen2.5-Math-7B) likelihoods and confidences for MATH500 responses.** Left: We plot the log-likelihoods (relative to the base model) of original, power sampling, and GRPO responses over MATH500. Right: We do the same but for confidences relative to the base model. We observe that GRPO samples from the highest likelihood and confidence regions with power sampling close behind, which correlates with higher empirical accuracy.

Solving Long Horizon Reasoning Problems

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Engineering at Anthropic



Building a C compiler with a team of parallel Claudes

Published Feb 05, 2026

We tasked Opus 4.6 using agent teams to build a C Compiler, and then (mostly) walked away. Here's what it taught us about the future of autonomous software development.

Reinforcement Learning for Reasoning in Large Language Models with *One* Training Example

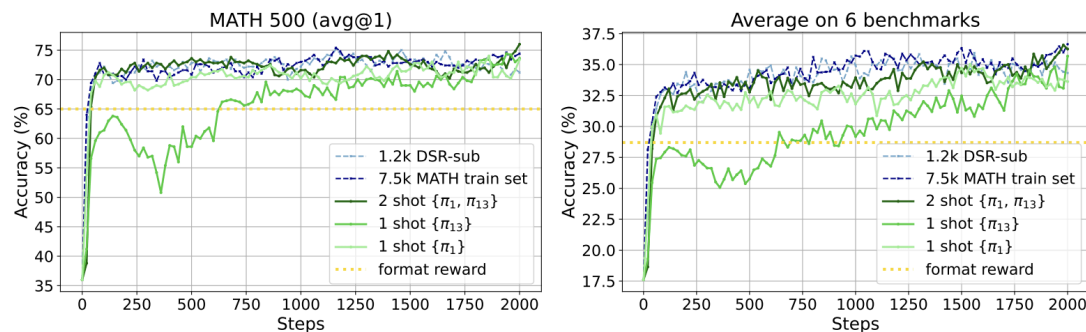


Figure 1: **RLVR with 1 example (green)** can perform as well as using datasets with **thousands of examples (blue)**. Left/Right corresponds to MATH500/Average performance on 6 mathematical reasoning benchmarks (MATH500, AIME24, AMC23, Minerva Math, OlympiadBench, and AIME25). Base model is Qwen2.5-Math-1.5B. π_1 and π_{13} are examples defined by Eqn. 2 and detailed in Tab. 2, and they are from the 1.2k DeepScalerR subset (DSR-sub). Setup details are in Sec. 3.1. We find that RLVR with 1 example $\{\pi_{13}\}$ (35.7%) performs close to that with 1.2k DSR-sub (35.9%), and RLVR with 2 examples $\{\pi_1, \pi_{13}\}$ (36.6%) even performs better than RLVR with DSR-sub and as well as using 7.5k MATH train dataset (36.7%). **Format reward (gold)** (Appendix C.2.3) serves as a baseline for format correction. Detailed results are in Appendix C.1.1. Additional results for non-mathematical reasoning tasks are in Tab. 1.

Resources

